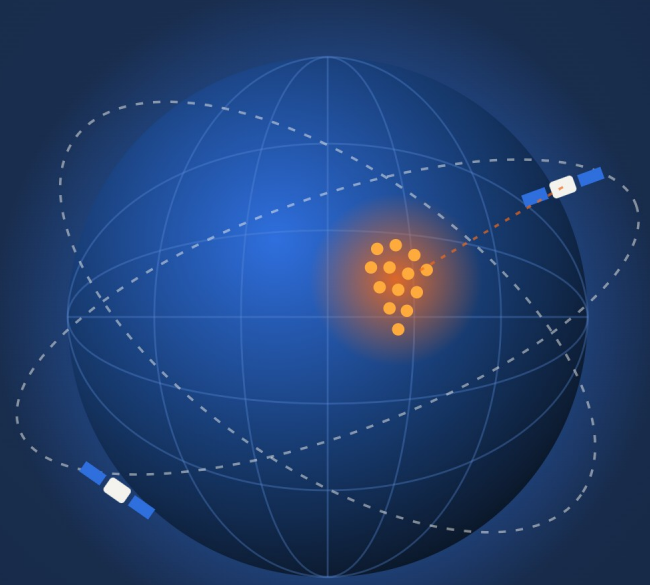
SATELLITE INTELLIGENCE FOR EVERY DISTRICT OF INDIA

Bhu-Drishti

Ask a question about floods, farm fires or new construction in plain words, and get a trustworthy answer from satellite data, with a report ready for officials.



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Satellite Data & AI Engine

Pramugdha Gupta
Frontend

Aditi
Backend & APIs

THE PROBLEM

Officials need answers in hours, not weeks

Disasters move fast

Floods drown crops and stubble fires spread in days. By the time a field survey is done, the moment to act has passed.

Data is scattered

Radar, optical and fire data sit with different agencies, each with its own keys, formats and limits.

Images aren’t answers

Turning imagery into “how much, where, how sure” needs experts most district offices do not have.

735 districts in India, each needing timely, evidence-backed information on floods, fires and land change.

HOW IT WORKS

From a plain question to a signed-off report

01

Ask

Type a question in your own words and pick a district.

02

Understand

The question becomes a precise plan: task, target, dates.

03

Sample

Smart sampling reads only the satellite tiles that matter.

04

Answer

A number with an honest range, a map and the evidence.

05

Report

Two PDFs, emailed to officials after review.

“Is farmland under water in Barpeta right now?”

“About [__]% of cropland is flooded (range [__]–[__]%)”

WHAT IT WATCHES

Four workflows, one platform

Flooded cropland

Radar sees through monsoon clouds to measure how much farmland is under water.

Farm fires

Every NASA fire hotspot counted and mapped, for stubble burning and wildfire watch.

Burned cropland + recovery

Damaged fields found, with practical advice for farmers to restore production.

New construction

New built-up area between two dates, for planning and encroachment checks.

REAL SATELLITE DATA

Six live sources behind one data layer

Unified data layer

- Keys stay on the server, never in the browser
- Every answer records its source
- Caching and retries built in
- District boundaries for all of India

Planetary Computer
Sentinel-1 radar, Sentinel-2 optical

Copernicus CDSE
European Sentinel processing

Google Earth Engine
AlphaEarth land embeddings

NASA FIRMS
Active fire hotspots

NASA Earthdata
Earth observation archives

Open-Meteo
Rainfall and weather context

THE AI ENGINE · AEQE

Accurate answers without reading every pixel



01 Smart sampling

Groups similar tiles and reads the most informative ones first.

02 Honest ranges, any time

Confidence ranges stay valid whenever you stop, and tighten as more tiles arrive.

03 Full count when cheap

Fire questions use every hotspot, with no estimate needed.

04 Says when it doesn’t know

Cloudy or missing data is reported as “not enough evidence”.

SYSTEM DESIGN

Four layers, each with one job

Web app

React and Vite, interactive maps, 3D globe, light and dark themes

API

Node and Fastify: accounts, monitors, alerts, PDF reports, audit log

AEQE engine

Python and FastAPI: question parsing, sampling, statistics

Data sources

Six satellite and weather providers, read on demand

SQLite or Postgres for users, projects and reports

Email and n8n automation deliver reports

Docker Compose runs it all with one command

THE PRODUCT

Built for district officials

Everything in one workspace

- Dashboard
- Explore
- Areas
- Monitors
- Alerts
- Reports
- Recipients
- History

Two reports for every answer

SUMMARY PDF

For everyone

Plain words, how sure we are, a map and what to do now

TECHNICAL PDF

For experts

Method, confidence ranges, data sources and evidence

Reports are emailed automatically; officials receive them after human approval.

TRUST AND SECURITY

Ready for real-world use

65
automated engine tests

18
automated backend tests

735
districts covered with live data

Login required, passwords hashed

Secure reset and sign-out everywhere

Security headers and rate limits

Human approval before reports go out

Daily backups with one-step restore

Tests run on every code change

Health checks and auto-restart

Audit log of every delivery

ROADMAP

What comes next

DONE · PHASE 1

Working platform

Live data, four workflows, monitors, alerts, two PDFs and email delivery

NEXT · LAUNCH

Go live

Cloud server with HTTPS, a professional email service and field-data checks for accuracy

LATER · PHASE 2

Reach further

WhatsApp and SMS alerts, and roles for state, district and block teams

Thank you

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